

12576 - Nisanth Saravanan

L1 MySQL Assessmet

Online Auction System

This case study involves the design of an online auction system that enables users to bid on items and win auctions. The system manages auction items, bids, and winner determination. The database schema ensures secure and transparent management of auction processes.

Task:

- Perform W3H analysis on the given Case Study.
- Design an ERD
- Create tables for an implemented ERD
- Add primary key, foreign key, and the necessary constraints.
- Insert data into the tables (min of 7 rows)
- Generate the schema diagram (Use reverse engineering)

Documents to be submitted in PDF file format.

- 1.W3H doc.
- 2.Output Screen doc. (ERD , Table Creation Queries , Table Data)
3. MYSQL Coding Screens.
4. And upload the Query File in Github and submit the URL.

GitHub Link:

[RelevantzRIX2024/Assessments/SQL at 12576 Nisanth · benhar-velankanni/RelevantzRIX2024](https://github.com/RelevantzRIX2024/Assessments/SQL-at-12576-Nisanth-benhar-velankanni/RelevantzRIX2024)

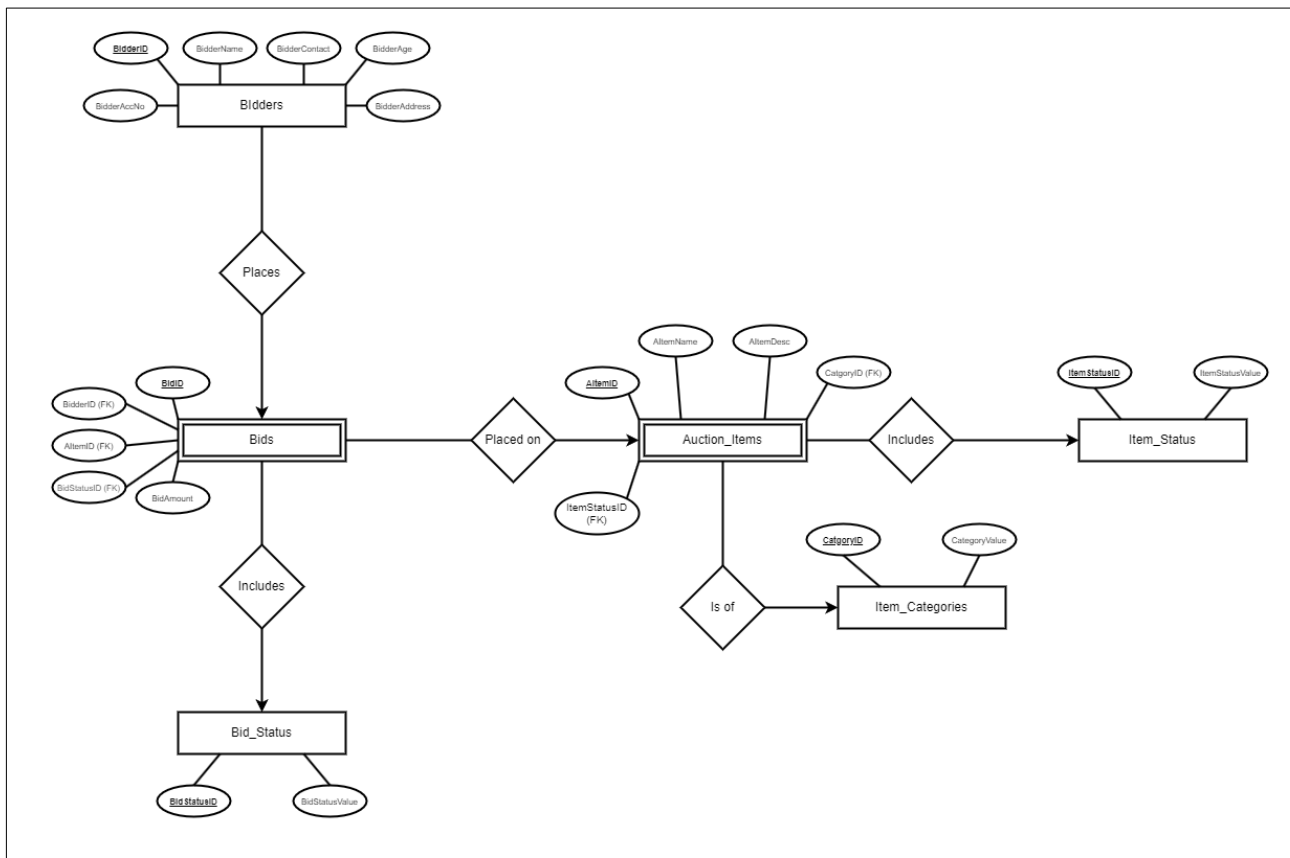
W3H Analysis

Project Title: Online Auction System	
What?	How?
1. What are the various entities needed for this database? A: Required Entities: a) Auction_Items b) Bidders c) Bids	1. Auction_Items: a) Store details of auction_items: Method 1: Store the details in a single table. Method 2: Divide the details into a few more tables, for example Categories as an individual relation, and make use of references. Method 3: Store the details with non-atomic values in columns. b) Store the status of each item: Method 1: Store the details in the same table. Method 2: Divide the details into a few more tables. Method 3: Store the details with non-atomic values in columns.
2. What are the necessary functionalities of Auction_Items? A: Auction_Items: a) Store details of auction items. b) Store the status of each item.	
3. What are the necessary functionalities of Bidders? A: Bidders: a) Store details of Bidders.	
4. What are the necessary functionalities of Bids? A: Bids: a) Store details of Bids. b) Store the status of each bid.	2. Bidders: a) Store the details of Bidders: Method 1: Store the details in the same table. Method 2: Divide the details into a few more tables. Method 3: Store the details with non-atomic values in columns.

	3. Bids: a) Store the details of Bids: Method 1: Store the details in the same table, using only the minimal amount of data to have a significant relation. Method 2: Divide the details into a few more tables. Method 3: Store the details with non-atomic values in columns. b) Store the status of each item: Method 1: Store the details in the same table. Method 2: Divide the details into a few more tables. Method 3: Store the details with non-atomic values in columns.
1. Auction_Items: a) Store details of auction_items: Method 2: Divide the details into a few more tables, for example Categories as an individual relation, and make use of references. A: Provides efficient relation with sufficient normalisation. b) Store the status of each item: Method 2: Divide the details into a few more tables. A: Provides efficient relation with sufficient normalisation. 2. Bidders: a) Store the details of Bidders: Method 1: Store the details in the same table.	1. Auction_Items: a) Store details of auction_items: The details can be stored in a single table, but will contain at least a single column of high redundancy. And non-atomic values are inefficient to our cause. b) Store the status of each item: The details can be stored in a single table, but will contain at least a single column of high redundancy. And non-atomic values are inefficient to our cause. 2. Bidders: a) Store the details of Bidders: The need for a new relation asks for more join type queries in the future and aggregated values are a mess.

A: Provides sufficient normalisation, without needing a new relation in the mix. 3. Bids: a) Store the details of Bids: Method 1: Store the details in the same table, using only the minimal amount of data to have a significant relation. A: Provides sufficient normalisation, without needing a new relation in the mix. b) Store the status of each item: Method 2: Divide the details into a few more tables. A: Provides efficient relation with sufficient normalisation.	3. Bids: a) Store the details of Bids: The need for a new relation asks for more join type quires in the future and aggregated values are a mess. b) Store the status of each item: The details can be stored in a single table, but will contain at least a single column of high redundancy. And non-atomic values are inefficient to our cause.
Why?	Why Not?

ER Diagram:



Auctions.sql

```
CREATE DATABASE AUCTION_HOUSE;  
USE AUCTION_HOUSE;
```

-- ITEM STATUS TABLE:

```
CREATE TABLE ITEM_STATUS (  
    ItemStatusID INT PRIMARY KEY,  
    ItemStatusValue VARCHAR(50) UNIQUE  
);
```

```
INSERT INTO ITEM_STATUS (ItemStatusID,ItemStatusValue)  
VALUES  
    (1, 'SOLD'),  
    (0, 'UNSOLD');
```

```
SELECT  
    ItemStatusID 'ID', ItemStatusValue 'STATUS'  
FROM  
    ITEM_STATUS;
```

Result Grid			Filter Rows:	Export: 	Wrap Cell Content: 
	ID	STATUS			
▶	1	SOLD			
	0	UNSOLD			

ITEM_STATUS 14 x

-- BID STATUS TABLE:

```
CREATE TABLE BID_STATUS(  
    BidStatusID INT PRIMARY KEY,  
    BidStatusValue VARCHAR(50) UNIQUE  
);
```

```
INSERT INTO BID_STATUS (BidStatusID, BidStatusValue)  
VALUES  
    (1, 'WINNER'),  
    (2, 'LOST'),  
    (3, 'PENDING');
```

```
SELECT  
    BidStatusID 'ID', BidStatusValue 'VALUE'  
FROM  
    BID_STATUS;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	ID	VALUE			
▶	2	LOST			
	3	PENDING			
	1	WINNER			

BID_STATUS 15 x

-- ITEM CATEGORIES TABLE:

```
CREATE TABLE ITEM_CATEGORIES(  
    CatgoryID INT PRIMARY KEY,  
    CategoryValue VARCHAR(50) UNIQUE  
);
```

```
INSERT INTO ITEM_CATEGORIES (CatgoryID, CategoryValue)
```

```
VALUES
```

```
    (100, 'ANTIQUES'),  
    (101, 'ELECTRONICS'),  
    (102, 'FASHION'),  
    (103, 'JEWELLERY'),  
    (104, 'ART'),  
    (105, 'PRE-HISTORIC'),  
    (106, 'HISTORIC'),  
    (107, 'ORIENTAL'),  
    (108, 'WAR MEMROBILLIA'),  
    (109, 'LITERATURE'),  
    (110, 'MISC');
```

```
SELECT
```

```
    CatgoryID 'ID', CategoryValue 'CATEGORY'
```

```
FROM
```

```
    ITEM_CATEGORIES;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	ID	CATEGORY			
▶	100	ANTIQUES			
	104	ART			
	101	ELECTRONICS			
	102	FASHION			
	106	HISTORIC			
	103	JEWELLERY			
	109	LITERATURE			
	110	MISC			
	107	ORIENTAL			
	105	PRE-HISTORIC			
	108	WAR MEMRO...			

ITEM_CATEGORIES 16 x

-- AUCTION ITEMS TABLE:

```
CREATE TABLE AUCTION_ITEMS(  
    AltemID INT AUTO_INCREMENT PRIMARY KEY,  
    AltemName VARCHAR(50),  
    AltemDesc VARCHAR(250),  
    CatgoryID INT,  
    ItemStatusID INT DEFAULT 0,  
    FOREIGN KEY (CatgoryID) REFERENCES ITEM_CATEGORIES(CatgoryID),  
    FOREIGN KEY (ItemStatusID) REFERENCES ITEM_STATUS(ItemStatusID)  
);
```

```
INSERT INTO AUCTION_ITEMS (AltemName, AltemDesc, CatgoryID)  
VALUES  
    ('TEMPLAR KNIGHT ARMOR', 'Lorem ipsum dolor sit amet',106),  
    ('THE OLDEST KNOWN GREEN TEA FROM THE SILK ROUTE', 'Lorem ipsum dolor sit amet',107),  
    ('MG42 - THE LMG', 'Lorem ipsum dolor sit amet',108),  
    ('THE FIRST COPY OF ANIMAL FARM', 'Lorem ipsum dolor sit amet',109),  
    ('A PARCHMENT OF OLD ROYAL DECREE', 'Lorem ipsum dolor sit amet',110),  
    ('COLLAR BONE OF A T-REX', 'Lorem ipsum dolor sit amet',105),  
    ('DANTE'S INFERNO', 'Lorem ipsum dolor sit amet',104),  
    ('KOHINOOR', 'Lorem ipsum dolor sit amet',103),  
    ('HITLER'S TRENCH COAT', 'Lorem ipsum dolor sit amet',102),  
    ('HUMMING BIRD VASE', 'Lorem ipsum dolor sit amet',100),  
    ('THE 1ST PHONE', 'Lorem ipsum dolor sit amet',101);
```

```
SELECT  
    A.AltemID 'ITEM ID',  
    A.AltemName 'NAME',  
    A.AltemDesc 'DESCRIPTION',  
    C.CategoryValue 'CATEGORY',  
    S.ItemStatusValue 'STATUS'  
FROM  
    AUCTION_ITEMS A  
    JOIN  
    ITEM_CATEGORIES C ON A.CatgoryID = C.CatgoryID  
    JOIN  
    ITEM_STATUS S ON A.ItemStatusID = S.ItemStatusID;
```

Result Grid					
Filter Rows:		Export:		Wrap Cell Content:	
ITEM ID	NAME	DESCRIPTION	CATEGORY	STATUS	
1	TEMPLAR KNIGHT ARMOR	Lorem ipsum dolor sit amet	HISTORIC	UNSOLD	
2	THE OLDEST KNOWN GREEN TEA FROM THE SILK ROUTE	Lorem ipsum dolor sit amet	ORIENTAL	UNSOLD	
3	MG42 - THE LMG	Lorem ipsum dolor sit amet	WAR MEMROBILLIA	UNSOLD	
4	THE FIRST COPY OF ANIMAL FARM	Lorem ipsum dolor sit amet	LITERATURE	UNSOLD	
5	A PARCHMENT OF OLD ROYAL DECREE	Lorem ipsum dolor sit amet	MISC	UNSOLD	
6	COLLAR BONE OF A T-REX	Lorem ipsum dolor sit amet	PRE-HISTORIC	UNSOLD	
7	DANTE'S INFERNO	Lorem ipsum dolor sit amet	ART	UNSOLD	
8	KOHINOOR	Lorem ipsum dolor sit amet	JEWELLERY	UNSOLD	
9	HITLER'S TRENCH COAT	Lorem ipsum dolor sit amet	FASHION	UNSOLD	
10	HUMMING BIRD VASE	Lorem ipsum dolor sit amet	ANTIQUES	UNSOLD	
11	THE 1ST PHONE	Lorem ipsum dolor sit amet	ELECTRONICS	UNSOLD	

-- BIDDERS TABLE:

```
CREATE TABLE BIDDERS(  
    BidderID INT AUTO_INCREMENT PRIMARY KEY,  
    BidderName VARCHAR(50),  
    BidderContact VARCHAR(10) NOT NULL,  
    BidderAge INT,  
    BidderAddress VARCHAR(250) NOT NULL,  
    BidderAccNo VARCHAR(10) NOT NULL UNIQUE,  
    CONSTRAINT CHK_BidderContact CHECK(BidderContact REGEXP '[0-9]{10}$'),  
    CONSTRAINT CHK_BidderAccNo CHECK(BidderAccNo REGEXP '[0-9]{10}$'),  
    CONSTRAINT CHK_BidderAge CHECK(BidderAge >= 18)  
);  
  
INSERT INTO BIDDERS (BidderName, BidderContact, BidderAge, BidderAddress, BidderAccNo)  
VALUES  
    ('JHON DOE', 1234567890, 23, 'Lorem ipsum dolor sit amet', 1234567890),  
    ('JANE DOE', 1245862350, 26, 'Lorem ipsum dolor sit amet', 1245862350),  
    ('JENNY DOE', 7541283609, 26, 'Lorem ipsum dolor sit amet', 7541283609),  
    ('JAKE DOE', 8965741238, 32, 'Lorem ipsum dolor sit amet', 8965741238),  
    ('MICKEY DOE', 8452367109, 67, 'Lorem ipsum dolor sit amet', 8452367109),  
    ('DUCKY DOE', 8541362079, 34, 'Lorem ipsum dolor sit amet', 2854136079),  
    ('MARY DOE', 1024586325, 87, 'Lorem ipsum dolor sit amet', 1024586325);  
  
SELECT  
    BidderID 'ID',  
    BidderName 'NAME',  
    BidderContact 'CONTACT',  
    BidderAge 'AGE',  
    BidderAddress 'ADDRESS',  
    BidderAccNo 'ACCOUNT NO'  
FROM  
    BIDDERS;
```

Result Grid		Filter Rows:		Export:	Wrap Cell Content:	
	ID	NAME	CONTACT	AGE	ADDRESS	ACCOUNT NO
▶	1	JHON DOE	1234567890	23	Lorem ipsum dolor sit amet	1234567890
	2	JANE DOE	1245862350	26	Lorem ipsum dolor sit amet	1245862350
	3	JENNY DOE	7541283609	26	Lorem ipsum dolor sit amet	7541283609
	4	JAKE DOE	8965741238	32	Lorem ipsum dolor sit amet	8965741238
	5	MICKEY DOE	8452367109	67	Lorem ipsum dolor sit amet	8452367109
	6	DUCKY DOE	8541362079	34	Lorem ipsum dolor sit amet	2854136079
	7	MARY DOE	1024586325	87	Lorem ipsum dolor sit amet	1024586325

-- BIDS TABLE:

```
CREATE TABLE BIDS(  
    BidID INT AUTO_INCREMENT PRIMARY KEY,  
    BidderID INT,  
    AltemID INT,  
    BidStatusID INT DEFAULT 3,  
    BidAmount INT,  
    FOREIGN KEY (BidderID) REFERENCES BIDDERS(BidderID),  
    FOREIGN KEY (AltemID) REFERENCES AUCTION_ITEMS(AltemID),  
    FOREIGN KEY (BidStatusID) REFERENCES BID_STATUS(BidStatusID)  
);
```

```
INSERT INTO BIDS (BidderID, AltemID, BidAmount)
```

VALUES

```
(1, 3, 50000),  
(3, 4, 45000),  
(7, 5, 75000),  
(2, 3, 84000),  
(1, 4, 96000),  
(4, 5, 98000),  
(6, 6, 75000),  
(7, 6, 784000);
```

SELECT

```
B.BidID 'BID ID',  
BD.BidderName 'BIDDER NAME',  
A.AltemName 'ITEM NAME',  
S.BidStatusValue 'STATUS',  
B.BidAmount 'BID AMOUNT'
```

FROM

BIDS B

JOIN

BIDDERS BD ON B.BidderID = BD.BidderID

JOIN

AUCTION_ITEMS A ON B.AltemID = A.AltemID

JOIN

BID_STATUS S ON B.BidStatusID = S.BidStatusID;

Result Grid					
		Filter Rows:		Export:	Wrap Cell Content: fA
	BID ID	BIDDER NAME	ITEM NAME	STATUS	BID AMOUNT
▶	1	JHON DOE	MG42 - THE LMG	PENDING	50000
	2	JENNY DOE	THE FIRST COPY OF ANIMAL FARM	PENDING	45000
	3	MARY DOE	A PARCHMENT OF OLD ROYAL DECREE	PENDING	75000
	4	JANE DOE	MG42 - THE LMG	PENDING	84000
	5	JHON DOE	THE FIRST COPY OF ANIMAL FARM	PENDING	96000
	6	JAKE DOE	A PARCHMENT OF OLD ROYAL DECREE	PENDING	98000
	7	DUCKY DOE	COLLAR BONE OF A T-REX	PENDING	75000
	8	MARY DOE	COLLAR BONE OF A T-REX	PENDING	784000

Result 21 ×

Updating values:

Before Updating:

```
SELECT
  B.BidID 'BID ID',
  BD.BidderName 'BIDDER NAME',
  A.AltemName 'ITEM NAME',
  S.BidStatusValue 'STATUS',
  B.BidAmount 'BID AMOUNT'
FROM
  BIDS B
  JOIN
  BIDDERS BD ON B.BidderID = BD.BidderID
  JOIN
  AUCTION_ITEMS A ON B.AltemID = A.AltemID
  JOIN
  BID_STATUS S ON B.BidStatusID = S.BidStatusID
WHERE
  B.AltemID = 5
ORDER BY B.BidAmount DESC;
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	BID ID	BIDDER NAME	ITEM NAME	STATUS	BID AMOUNT
▶	6	JAKE DOE	A PARCHMENT OF OLD ROYAL DECREE	PENDING	98000
	3	MARY DOE	A PARCHMENT OF OLD ROYAL DECREE	PENDING	75000

Result 29 x

After Updating:

-- Updating the bid status of a particular bid:

```
SELECT
    B.BidID 'BID ID',
    BD.BidderName 'BIDDER NAME',
    A.AltemName 'ITEM NAME',
    S.BidStatusValue 'STATUS',
    B.BidAmount 'BID AMOUNT'
FROM
    BIDS B
    JOIN
        BIDDERS BD ON B.BidderID = BD.BidderID
    JOIN
        AUCTION_ITEMS A ON B.AltemID = A.AltemID
    JOIN
        BID_STATUS S ON B.BidStatusID = S.BidStatusID
WHERE
    B.AltemID = 5
ORDER BY B.BidAmount DESC;
```

```
UPDATE BIDS
SET BidStatusID = 1
WHERE (AltemID = 5) AND (BidAmount = 98000);
```

```
UPDATE BIDS
SET BidStatusID = 2
WHERE (AltemID = 5) AND (BidAmount = 75000);
```

	BID ID	BIDDER NAME	ITEM NAME	STATUS	BID AMOUNT
▶	6	JAKE DOE	A PARCHMENT OF OLD ROYAL DECREE	WINNER	98000
	3	MARY DOE	A PARCHMENT OF OLD ROYAL DECREE	LOST	75000

-- Updating the Item Status after sale:

UPDATE AUCTION_ITEMS

SET

ItemStatusID = 1

WHERE

AltemID = 5;

Before:

	ITEM ID	NAME	DESCRIPTION	CATEGORY	STATUS
▶	1	TEMPLAR KNIGHT ARMOR	Lorem ipsum dolor sit amet	HISTORIC	UNSOLD
	2	THE OLDEST KNOWN GREEN TEA FROM THE SILK ROUTE	Lorem ipsum dolor sit amet	ORIENTAL	UNSOLD
	3	MG42 - THE LMG	Lorem ipsum dolor sit amet	WAR MEMROBILLIA	UNSOLD
	4	THE FIRST COPY OF ANIMAL FARM	Lorem ipsum dolor sit amet	LITERATURE	UNSOLD
	5	A PARCHMENT OF OLD ROYAL DECREE	Lorem ipsum dolor sit amet	MISC	UNSOLD
	6	COLLAR BONE OF A T-REX	Lorem ipsum dolor sit amet	PRE-HISTORIC	UNSOLD
	7	DANTE'S INFERNO	Lorem ipsum dolor sit amet	ART	UNSOLD
	8	KOHINOOR	Lorem ipsum dolor sit amet	JEWELLERY	UNSOLD
	9	HITLER'S TRENCH COAT	Lorem ipsum dolor sit amet	FASHION	UNSOLD
	10	HUMMING BIRD VASE	Lorem ipsum dolor sit amet	ANTIQUES	UNSOLD
	11	THE 1ST PHONE	Lorem ipsum dolor sit amet	ELECTRONICS	UNSOLD

Result 82 ×

After:

Result Grid



 Filter Rows:

Export:



Wrap Cell Content:



	ITEM ID	NAME	DESCRIPTION	CATEGORY	STATUS
▶	1	TEMPLAR KNIGHT ARMOR	Lorem ipsum dolor sit amet	HISTORIC	UNSOLD
	2	THE OLDEST KNOWN GREEN TEA FROM THE SILK ROUTE	Lorem ipsum dolor sit amet	ORIENTAL	UNSOLD
	3	MG42 - THE LMG	Lorem ipsum dolor sit amet	WAR MEMROBILLIA	UNSOLD
	4	THE FIRST COPY OF ANIMAL FARM	Lorem ipsum dolor sit amet	LITERATURE	UNSOLD
	5	A PARCHMENT OF OLD ROYAL DECREE	Lorem ipsum dolor sit amet	MISC	SOLD
	6	COLLAR BONE OF A T-REX	Lorem ipsum dolor sit amet	PRE-HISTORIC	UNSOLD
	7	DANTE'S INFERNO	Lorem ipsum dolor sit amet	ART	UNSOLD
	8	KOHINOOR	Lorem ipsum dolor sit amet	JEWELLERY	UNSOLD
	9	HITLER'S TRENCH COAT	Lorem ipsum dolor sit amet	FASHION	UNSOLD
	10	HUMMING BIRD VASE	Lorem ipsum dolor sit amet	ANTIQUES	UNSOLD
	11	THE 1ST PHONE	Lorem ipsum dolor sit amet	ELECTRONICS	UNSOLD

Result 83 ×

Deletion of rows:

Deletion of Concluded Bids and Deleting sold items:

Before:

Result Grid





Filter Rows:

Export:



Wrap Cell Content:



	ITEM ID	NAME	DESCRIPTION	CATEGORY	STATUS
▶	1	TEMPLAR KNIGHT ARMOR	Lorem ipsum dolor sit amet	HISTORIC	UNSOLD
	2	THE OLDEST KNOWN GREEN TEA FROM THE SILK ROUTE	Lorem ipsum dolor sit amet	ORIENTAL	UNSOLD
	3	MG42 - THE LMG	Lorem ipsum dolor sit amet	WAR MEMROBILLIA	UNSOLD
	4	THE FIRST COPY OF ANIMAL FARM	Lorem ipsum dolor sit amet	LITERATURE	UNSOLD
	5	A PARCHMENT OF OLD ROYAL DECREE	Lorem ipsum dolor sit amet	MISC	SOLD
	6	COLLAR BONE OF A T-REX	Lorem ipsum dolor sit amet	PRE-HISTORIC	UNSOLD
	7	DANTE'S INFERNO	Lorem ipsum dolor sit amet	ART	UNSOLD
	8	KOHINOOR	Lorem ipsum dolor sit amet	JEWELLERY	UNSOLD
	9	HITLER'S TRENCH COAT	Lorem ipsum dolor sit amet	FASHION	UNSOLD
	10	HUMMING BIRD VASE	Lorem ipsum dolor sit amet	ANTIQUES	UNSOLD
	11	THE 1ST PHONE	Lorem ipsum dolor sit amet	ELECTRONICS	UNSOLD

Result Grid




Filter Rows:

Export: 

Wrap Cell Content: 

	BID ID	BIDDER NAME	ITEM NAME	STATUS	BID AMOUNT
▶	1	JHON DOE	MG42 - THE LMG	PENDING	50000
	2	JENNY DOE	THE FIRST COPY OF ANIMAL FARM	PENDING	45000
	3	MARY DOE	A PARCHMENT OF OLD ROYAL DECREE	LOST	75000
	4	JANE DOE	MG42 - THE LMG	PENDING	84000
	5	JHON DOE	THE FIRST COPY OF ANIMAL FARM	PENDING	96000
	6	JAKE DOE	A PARCHMENT OF OLD ROYAL DECREE	WINNER	98000
	7	DUCKY DOE	COLLAR BONE OF A T-REX	PENDING	75000
	8	MARY DOE	COLLAR BONE OF A T-REX	PENDING	784000

After:

-- Deletion of sold items:

```
DELETE FROM AUCTION_ITEMS
```

```
WHERE
```

```
ItemStatusID = 1;
```

-- Deletion of Concluded Bids:

```
DELETE FROM BIDS
```

```
WHERE
```

```
BidStatusID != 3;
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	BID ID	BIDDER NAME	ITEM NAME	STATUS	BID AMOUNT
▶	1	JHON DOE	MG42 - THE LMG	PENDING	50000
	2	JENNY DOE	THE FIRST COPY OF ANIMAL FARM	PENDING	45000
	4	JANE DOE	MG42 - THE LMG	PENDING	84000
	5	JHON DOE	THE FIRST COPY OF ANIMAL FARM	PENDING	96000
	7	DUCKY DOE	COLLAR BONE OF A T-REX	PENDING	75000
	8	MARY DOE	COLLAR BONE OF A T-REX	PENDING	784000

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	ITEM ID	NAME	DESCRIPTION	CATEGORY	STATUS
▶	1	TEMPLAR KNIGHT ARMOR	Lorem ipsum dolor sit amet	HISTORIC	UNSOLD
	2	THE OLDEST KNOWN GREEN TEA FROM THE SILK ROUTE	Lorem ipsum dolor sit amet	ORIENTAL	UNSOLD
	3	MG42 - THE LMG	Lorem ipsum dolor sit amet	WAR MEMROBILLIA	UNSOLD
	4	THE FIRST COPY OF ANIMAL FARM	Lorem ipsum dolor sit amet	LITERATURE	UNSOLD
	6	COLLAR BONE OF A T-REX	Lorem ipsum dolor sit amet	PRE-HISTORIC	UNSOLD
	7	DANTE'S INFERNO	Lorem ipsum dolor sit amet	ART	UNSOLD
	8	KOHINOOR	Lorem ipsum dolor sit amet	JEWELLERY	UNSOLD
	9	HITLER'S TRENCH COAT	Lorem ipsum dolor sit amet	FASHION	UNSOLD
	10	HUMMING BIRD VASE	Lorem ipsum dolor sit amet	ANTIQUES	UNSOLD
	11	THE 1ST PHONE	Lorem ipsum dolor sit amet	ELECTRONICS	UNSOLD

(Only the rows with sold status and rows without pending status were deleted.)

Store Procedures:

-- Store Procedure to See if an item exists:

DELIMITER //

CREATE PROCEDURE spSearchItem (IN ITEM_NAME VARCHAR(50), OUT RESULT VARCHAR(50))

BEGIN

 DECLARE COUNTER INT;

SELECT

 COUNT(*)

INTO COUNTER FROM

 AUCTION_ITEMS

WHERE

 ITEM_NAME = AltemName;

 IF (COUNTER > 0) THEN

 SET RESULT = 'ITEM EXISTS!';

 ELSE SET RESULT = "ITEM DOESN'T EXIST!";

 END IF;

END //

DELIMITER ;

CALL spSearchItem('TEMPLAR KNIGHT ARMOR', @result);

SELECT @result;

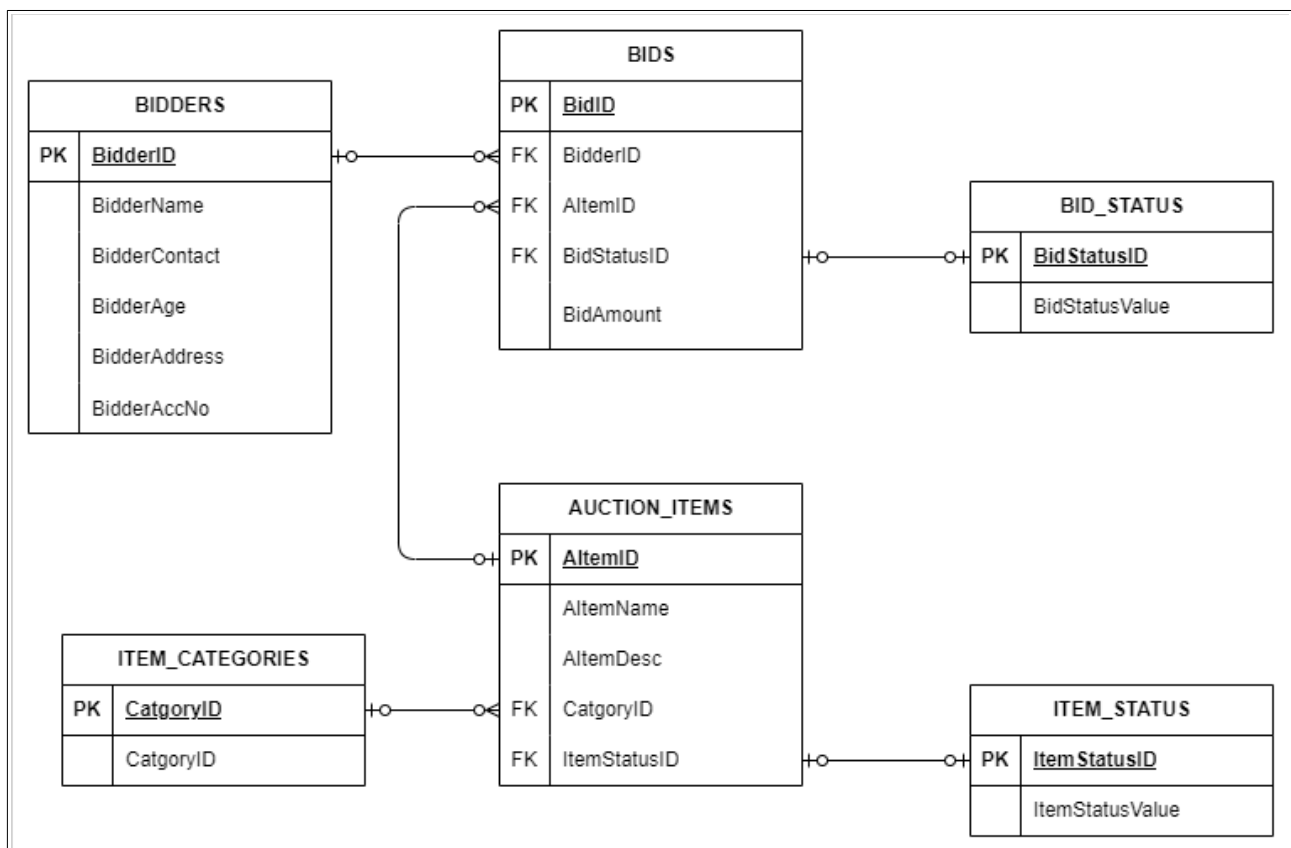
Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	@result			
▶	ITEM EXISTS!			

CALL spSearchItem('RANDOM GARBAGE', @result);

SELECT @result;

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	@result			
▶	ITEM DOESN'T EXIST!			

Schema Diagram:



Complete SQL Code:

```
CREATE DATABASE AUCTION_HOUSE;
```

```
--
```

```
USE AUCTION_HOUSE;
```

```
--
```

```
-- ITEM STATUS TABLE:
```

```
CREATE TABLE ITEM_STATUS (  
    ItemStatusID INT PRIMARY KEY,  
    ItemStatusValue VARCHAR(50) UNIQUE  
);
```

```
INSERT INTO ITEM_STATUS (ItemStatusID,ItemStatusValue)  
VALUES  
    (1, 'SOLD'),  
    (0, 'UNSOLD');
```

```
SELECT  
    ItemStatusID 'ID', ItemStatusValue 'STATUS'  
FROM  
    ITEM_STATUS;
```

```
--
```

```
-- BID STATUS TABLE:
```

```
CREATE TABLE BID_STATUS(  
    BidStatusID INT PRIMARY KEY,  
    BidStatusValue VARCHAR(50) UNIQUE  
);
```

```
INSERT INTO BID_STATUS (BidStatusID, BidStatusValue)  
VALUES
```

```
    (1, 'WINNER'),
    (2, 'LOST'),
    (3, 'PENDING');
```

```
SELECT
    BidStatusID 'ID', BidStatusValue 'VALUE'
FROM
    BID_STATUS;
```

```
--
```

```
-- ITEM CATEGORIES TABLE:
CREATE TABLE ITEM_CATEGORIES(
    CatgoryID INT PRIMARY KEY,
    CategoryValue VARCHAR(50) UNIQUE
);
```

```
INSERT INTO ITEM_CATEGORIES (CatgoryID, CategoryValue)
VALUES
    (100, 'ANTIQUES'),
    (101, 'ELECTRONICS'),
    (102, 'FASHION'),
    (103, 'JEWELLERY'),
    (104, 'ART'),
    (105, 'PRE-HISTORIC'),
    (106, 'HISTORIC'),
    (107, 'ORIENTAL'),
    (108, 'WAR MEMROBILLIA'),
    (109, 'LITERATURE'),
    (110, 'MISC');
```

```
SELECT
    CatgoryID 'ID', CategoryValue 'CATEGORY'
FROM
    ITEM_CATEGORIES;
```

```
--
```

```
-- AUCTION ITEMS TABLE:
```

```

CREATE TABLE AUCTION_ITEMS(
    AItemID INT AUTO_INCREMENT PRIMARY KEY,
    AItemName VARCHAR(50),
    AItemDesc VARCHAR(250),
    CatgoryID INT,
    ItemStatusID INT DEFAULT 0,
    FOREIGN KEY (CatgoryID) REFERENCES ITEM_CATEGORIES(CatgoryID),
    FOREIGN KEY (ItemStatusID) REFERENCES ITEM_STATUS(ItemStatusID)
);

```

```

INSERT INTO AUCTION_ITEMS (AItemName, AItemDesc, CatgoryID)
VALUES
    ('TEMPLAR KNIGHT ARMOR', 'Lorem ipsum dolor sit amet',106),
    ('THE OLDEST KNOWN GREEN TEA FROM THE SILK ROUTE', 'Lorem ipsum dolor sit
amet',107),
    ('MG42 - THE LMG', 'Lorem ipsum dolor sit amet',108),
    ('THE FIRST COPY OF ANIMAL FARM', 'Lorem ipsum dolor sit amet',109),
    ('A PARCHMENT OF OLD ROYAL DECREE', 'Lorem ipsum dolor sit amet',110),
    ('COLLAR BONE OF A T-REX', 'Lorem ipsum dolor sit amet',105),
    ("DANTE'S INFERNO", 'Lorem ipsum dolor sit amet',104),
    ('KOHINOOR', 'Lorem ipsum dolor sit amet',103),
    ("HITLER'S TRENCH COAT", 'Lorem ipsum dolor sit amet',102),
    ('HUMMING BIRD VASE', 'Lorem ipsum dolor sit amet',100),
    ('THE 1ST PHONE', 'Lorem ipsum dolor sit amet',101);

```

```

SELECT
    A.AItemID 'ITEM ID',
    A.AItemName 'NAME',
    A.AItemDesc 'DESCRIPTION',
    C.CategoryValue 'CATEGORY',
    S.ItemStatusValue 'STATUS'
FROM
    AUCTION_ITEMS A
    JOIN
    ITEM_CATEGORIES C ON A.CatgoryID = C.CatgoryID
    JOIN
    ITEM_STATUS S ON A.ItemStatusID = S.ItemStatusID;

```

--

-- BIDDERS TABLE:

```

CREATE TABLE BIDDERS(
    BidderID INT AUTO_INCREMENT PRIMARY KEY,
    BidderName VARCHAR(50),
    BidderContact VARCHAR(10) NOT NULL,
    BidderAge INT,
    BidderAddress VARCHAR(250) NOT NULL,
    BidderAccNo VARCHAR(10) NOT NULL UNIQUE,
    CONSTRAINT CHK_BidderContact CHECK(BidderContact REGEXP '[0-9]{10}$'),
    CONSTRAINT CHK_BidderAccNo CHECK(BidderAccNo REGEXP '[0-9]{10}$'),
    CONSTRAINT CHK_BidderAge CHECK(BidderAge >= 18)
);

```

```

INSERT INTO BIDDERS (BidderName, BidderContact, BidderAge, BidderAddress, BidderAccNo)
VALUES
    ('JHON DOE', 1234567890, 23, 'Lorem ipsum dolor sit amet', 1234567890),
    ('JANE DOE', 1245862350, 26, 'Lorem ipsum dolor sit amet', 1245862350),
    ('JENNY DOE', 7541283609, 26, 'Lorem ipsum dolor sit amet', 7541283609),
    ('JAKE DOE', 8965741238, 32, 'Lorem ipsum dolor sit amet', 8965741238),
    ('MICKEY DOE', 8452367109, 67, 'Lorem ipsum dolor sit amet', 8452367109),
    ('DUCKY DOE', 8541362079, 34, 'Lorem ipsum dolor sit amet', 2854136079),
    ('MARY DOE', 1024586325, 87, 'Lorem ipsum dolor sit amet', 1024586325);

```

```

SELECT
    BidderID 'ID',
    BidderName 'NAME',
    BidderContact 'CONTACT',
    BidderAge 'AGE',
    BidderAddress 'ADDRESS',
    BidderAccNo 'ACCOUNT NO'
FROM
    BIDDERS;

```

--

-- BIDS TABLE:

```

CREATE TABLE BIDS(
    BidID INT AUTO_INCREMENT PRIMARY KEY,
    BidderID INT,
    AltemID INT,
    BidStatusID INT DEFAULT 3,
    BidAmount INT,

```

```
FOREIGN KEY (BidderID) REFERENCES BIDDERS(BidderID),
FOREIGN KEY (AltemID) REFERENCES AUCTION_ITEMS(AltemID),
FOREIGN KEY (BidStatusID) REFERENCES BID_STATUS(BidStatusID)
);
```

```
INSERT INTO BIDS (BidderID, AltemID, BidAmount)
VALUES
    (1, 3, 50000),
    (3, 4, 45000),
    (7, 5, 75000),
    (2, 3, 84000),
    (1, 4, 96000),
    (4, 5, 98000),
    (6, 6, 75000),
    (7, 6, 784000);
```

```
SELECT
    B.BidID 'BID ID',
    BD.BidderName 'BIDDER NAME',
    A.AltemName 'ITEM NAME',
    S.BidStatusValue 'STATUS',
    B.BidAmount 'BID AMOUNT'
FROM
    BIDS B
    JOIN
    BIDDERS BD ON B.BidderID = BD.BidderID
    JOIN
    AUCTION_ITEMS A ON B.AltemID = A.AltemID
    JOIN
    BID_STATUS S ON B.BidStatusID = S.BidStatusID;
```

--

-- Updating the bid status of a particular bid:

```
SELECT
    B.BidID 'BID ID',
    BD.BidderName 'BIDDER NAME',
    A.AltemName 'ITEM NAME',
    S.BidStatusValue 'STATUS',
    B.BidAmount 'BID AMOUNT'
FROM
```

```
BIDS B
  JOIN
BIDDERS BD ON B.BidderID = BD.BidderID
  JOIN
AUCTION_ITEMS A ON B.AltemID = A.AltemID
  JOIN
  BID_STATUS S ON B.BidStatusID = S.BidStatusID
WHERE
  B.AltemID = 5
ORDER BY B.BidAmount DESC;
```

```
UPDATE BIDS
SET
  BidStatusID = 1
WHERE
  (AltemID = 5) AND (BidAmount = 98000);
```

```
UPDATE BIDS
SET
  BidStatusID = 2
WHERE
  (AltemID = 5) AND (BidAmount = 75000);
```

```
--
```

```
-- Updating the Item Status after sale:
```

```
UPDATE AUCTION_ITEMS
SET
  ItemStatusID = 1
WHERE
  AltemID = 5;
```

```
--
```

```
-- Deletion of sold items:
```

```
DELETE FROM AUCTION_ITEMS
WHERE
  ItemStatusID = 1;
```

-- Deletion of Concluded Bids:

DELETE FROM BIDS

WHERE

BidStatusID != 3;

--

-- Store Procedure to See if an item exists:

DELIMITER //

CREATE PROCEDURE spSearchItem (IN ITEM_NAME VARCHAR(50), OUT RESULT VARCHAR(50))

BEGIN

DECLARE COUNTER INT;

SELECT

COUNT(*)

INTO COUNTER FROM

AUCTION_ITEMS

WHERE

ITEM_NAME = AltemName;

IF (COUNTER > 0) THEN

SET RESULT = 'ITEM EXISTS!';

ELSE SET RESULT = "ITEM DOESN'T EXIST!";

END IF;

END //

DELIMITER ;

CALL spSearchItem('TEMPLAR KNIGHT ARMOR', @result);

SELECT @result;

CALL spSearchItem('RANDOM GARBAGE', @result);

SELECT @result;

--
